



Part 2 Overview: Building the EOM from First Principles

- ▶ Goal: derive 6 coupled first-order ODEs for $[r, v, \gamma, \psi, \phi, \lambda]$
- ▶ Approach: Newton's 2nd Law in ECI, expressed in vehicle-pointing coordinates
- ▶ Every term tracked explicitly — Coriolis, centripetal, planet curvature
- ▶ Derivation structure:
 - • Section 2.1: Setup and 7 key assumptions (~8 min)
 - • Section 2.2: Newton's law in rotating frame (~5 min)
 - • Section 2.3: 3 kinematic equations from geometry (~12 min)
 - • Section 2.4: Aerodynamic force decomposition (~8 min)
 - • Section 2.5: 3 force equations $\dot{v}$, $\dot{\gamma}$, $\dot{\psi}$ (~15 min)
 - • Section 2.6: Complete system + simplifications (~12 min)
- ▶ Primary source: Lesson 03-04 — Reentry EOM Development (Hicks Ch. 3)



The 7 Key Assumptions

- ▶ Hicks states these 7 assumptions explicitly before the EOM
- ▶ 1. Point mass — entry vehicle treated as a particle (no rotation)
- ▶ 2. Planetary rotation is constant about the polar axis
 - • $\omega_E = 7.2921 \times 10^{-5}$ rad/s (constant, not time-varying)
- ▶ 3. Planet-fixed and inertial frames are aligned at $t = 0$
 - • Sets the initial condition for the rotating-frame transformation
- ▶ 4. Constant mass — no fuel burn or ablation mass loss during entry
 - • Good assumption for unpowered entry; ablation effect is second order
- ▶ 5. Drag force acts in the direction opposite to the velocity vector
- ▶ 6. Lift force is perpendicular to the velocity vector
 - • Bank angle β then rotates lift about the velocity axis
- ▶ 7. Gravitational force is directed along the vector from point mass to planetary center
 - • Equivalent to: spherical, non-oblate Earth — no J_2 tangential component

[IMAGE]

Apollo CM heat shield after lunar return — ablated char layer visible
Caption: 'Assumption 4: constant mass. Apollo CM lost ~15 kg to ablation. How wrong are we?'

NASA JSC archive — public domain (search: Apollo CM heat shield AS12 or AS11)

TPS MOMENT | Each assumption is a test condition boundary. For a first crewed reentry, which of the 7 assumptions is most likely to be violated — and what does that do to your trajectory prediction? Classify the risk: Likely / Possible / Remote.



Setting Up Newton's 2nd Law in a Rotating Frame

- ▶ In ECI (inertial): $m \vec{a}_I = \vec{F}_{\text{gravity}} + \vec{F}_{\text{aero}}$
- ▶ Vehicle position in ECI: $\vec{r} = r \hat{r}(\varphi, \lambda, t)$
- ▶ Acceleration in ECI expressed via vehicle-pointing unit vectors:
 - $\vec{r} = [\ddot{r} - r \dot{\varphi}^2 - r \dot{\lambda}^2 \cos^2 \varphi] \hat{r} + [r \ddot{\varphi} + 2\dot{r} \dot{\varphi} - r \dot{\lambda}^2 \sin \varphi \cos \varphi] \hat{e}_\varphi + [r \cos \varphi \ddot{\lambda} + 2\dot{r} \dot{\lambda} \cos \varphi - 2r \dot{\varphi} \dot{\lambda} \sin \varphi] \hat{e}_\lambda$
- ▶ Key step: velocity in terms of state variables:
 - $\dot{r} = v \sin \gamma$
 - $r \dot{\varphi} = v \cos \gamma \cos \psi$
 - $r \dot{\lambda} \cos \varphi = v \cos \gamma \sin \psi$
- ▶ These three relations ARE the kinematic equations (next slides)
- ▶ Force vector: $\vec{F} = (-\mu/r^2) \hat{r} + [-D \hat{x}_w + L \hat{n}_L]$ expressed in vehicle-pointing
- ▶ Projection of acceleration onto $\hat{r}$, $\hat{e}_\varphi$, $\hat{e}_\lambda$ gives the three force equations



Kinematic Equations: Overview & Approach

- ▶ Three kinematic equations come from pure geometry — no forces needed
- ▶ They relate the rates of change of position to velocity:
 - $\dot{r}$: how fast does altitude change? → resolved along $\hat{r}$
 - $\dot{\phi}$: how fast does latitude change? → northward component of velocity
 - $\dot{\lambda}$: how fast does longitude change? → eastward component of velocity
- ▶ Velocity components in vehicle-pointing frame:
 - Along $\hat{r}$ (radial): $v_r = v \sin\gamma$
 - Along $\hat{e}_\phi$ (northward): $v_n = v \cos\gamma \cos\psi$
 - Along $\hat{e}_\lambda$ (eastward): $v_e = v \cos\gamma \sin\psi$
- ▶ These three projections directly give the three kinematic ODEs
- ▶ Note: these involve NO approximations — exact for spherical Earth geometry
- ▶ Source: Lesson 03-04



Kinematic EOM #1: $\dot{r} = v \sin \gamma$ (Altitude Rate)

- ▶ Physical meaning: rate of change of radial distance
 - $\dot{h} = \dot{r} = v \sin \gamma$ since $h = r - R_E$ and $\dot{R}_E = 0$
- ▶ Derivation: project velocity vector onto $\hat{r}$:
 - $\vec{v} \cdot \hat{r} = v \sin \gamma$ (γ is the FPA measured from local horizontal)
 - When $\gamma < 0$ (descending): $\dot{r} < 0$ ✓
 - When $\gamma = 0$ (horizontal flight): $\dot{r} = 0$ ✓
- ▶ Sign convention:
 - γ positive above local horizontal (climbing)
 - γ negative below local horizontal (descending)
 - Entry begins at $\gamma_E < 0$ typically between -1° and -20°
- ▶ Entry interface altitude: $h_{EI} \approx 120$ km (where aero forces become significant)

$$\dot{r} = v \sin \gamma$$



Kinematic EOM #2: $\dot{\phi}$ (Latitude Rate)

- ▶ Physical meaning: rate of change of geodetic latitude
- ▶ Velocity component northward (along $\hat{e}_\phi$): $v \cos\gamma \cos\psi$
 - $\psi = 0$ (heading north): $v_n = v \cos\gamma \rightarrow$ maximum northward progress
 - $\psi = 90^\circ$ (heading east): $v_n = 0 \rightarrow$ no latitude change
- ▶ Arc length along meridian = $r \phi$, so:
 - $r \dot{\phi} = v \cos\gamma \cos\psi$
 - $\dot{\phi} = (v \cos\gamma \cos\psi) / r$
- ▶ Sign convention:
 - $\dot{\phi} > 0$: moving toward north pole
 - $\dot{\phi} < 0$: moving toward south pole
- ▶ This equation is one of three that govern the geographic ground track

$$\dot{\phi} = (v \cos\gamma \cos\psi) / r$$



Kinematic EOM #3: $\dot{\lambda}$ (Longitude Rate)

- ▶ Physical meaning: rate of change of geographic longitude
- ▶ Velocity component eastward (along $\hat{e}_\lambda$): $v \cos\gamma \sin\psi$
- ▶ Arc length along parallel at latitude $\phi = r \cos\phi$, so:
 - $r \cos\phi \dot{\lambda} = v \cos\gamma \sin\psi$
 - $\dot{\lambda} = (v \cos\gamma \sin\psi) / (r \cos\phi)$
- ▶ Notes:
 - Denominator $\cos\phi \rightarrow \dot{\lambda}$ grows toward $\pm\infty$ at poles (pole crossing singularity)
 - For most reentry trajectories: $|\phi| < 70^\circ$, so $\cos\phi > 0.34$ — no issue
 - Eastward heading ($\psi = 90^\circ$): $\dot{\lambda} = v \cos\gamma / (r \cos\phi)$ — maximum east rate
- ▶ All three kinematic equations derived. They depend only on the state $[r, v, \gamma, \psi, \phi, \lambda]$
- ▶ No forces appear — these are pure geometric relations

$$\dot{\lambda} = (v \cos\gamma \sin\psi) / (r \cos\phi)$$



Kinematic Equations Summary: 3 ODEs from Geometry Alone

- ▶ The three kinematic equations — exact, no approximations:
- ▶ Read $\dot{r} = v \sin \gamma$
- ▶ Note: $\dot{\phi} = (v \cos \gamma \cos \psi) / r$
- • Read $\dot{\lambda} = (v \cos \gamma \sin \psi) / (r \cos \phi)$
- • The
- • For
- • This correction is small but important at lower altitudes and high latitudes
- ▶ Source: Lesson 03-04 — Reentry EOM Development



Force Equations: Projection Strategy

- ▶ Newton's 2nd Law: $m \vec{a} = \vec{F}_{\text{gravity}} + \vec{F}_{\text{drag}} + \vec{F}_{\text{lift}}$
- ▶ Strategy: express all forces in vehicle-pointing frame, then equate components
- ▶ Velocity in vehicle-pointing frame:
 - $\vec{v} = v \sin\gamma \hat{r} + v \cos\gamma \cos\psi \hat{e}_{\phi} + v \cos\gamma \sin\psi \hat{e}_{\lambda}$
- ▶ Time derivative of $\vec{v}$ in rotating vehicle-pointing frame introduces extra terms:
 - $d\vec{v}/dt = \dot{v} \hat{v} + v [\dot{\gamma} (\text{unit vector along FPA rotation}) + \dot{\psi} (\text{heading rotation})]$
- ▶ Projecting the equation of motion onto $\hat{v}$, the normal-to- v in vertical plane, and $\hat{y}_w$:
 - Along velocity $\hat{v}$: gives $\dot{v}$ equation
 - Normal (vertical plane): gives $\dot{\gamma}$ equation
 - Out-of-plane: gives $\dot{\psi}$ equation
- ▶ Three force equations result — one for each velocity-frame axis
- ▶ Source: Lesson 03-04



Aerodynamic Force Decomposition & Bank Angle β

- ▶ Drag vector: $\vec{F}_D = -D \hat{v}$ (along negative velocity direction)
- ▶ Lift vector (unbanked, $\beta = 0$): $\vec{F}_L = L \hat{n}$ where $\hat{n} \perp \hat{v}$ in vertical plane
- ▶ With bank angle β : lift rotates about velocity axis
 - Lift component in vertical plane: $L \cos\beta$
 - Lift component out of vertical plane (side force): $L \sin\beta$
- ▶ In vehicle-pointing frame components:
 - Radial ($\hat{r}$): $F_{\text{aero},r} = -D \sin\gamma + L \cos\beta \cos\gamma$
 - Northward: $F_{\text{aero},\phi} = -D \cos\gamma \cos\psi - L \cos\beta \sin\gamma \cos\psi - L \sin\beta \sin\gamma$
 - Eastward: $F_{\text{aero},\lambda} = -D \cos\gamma \sin\psi - L \cos\beta \sin\gamma \sin\psi + L \sin\beta \cos\gamma$
- ▶ Bank angle β is the primary guidance control variable in liftir
 - $\beta = 0^\circ$: lift directed away from Earth (maximum altitude rate)
 - $\beta = 180^\circ$: lift directed toward Earth (maximum sink rate)
- ▶ Source: Lesson 06 — Basic Planar Entry Equations

[DIAGRAM]

Capsule silhouette with force vectors:
D along -velocity (red arrow)
L perpendicular in vertical plane (blue arrow, $\beta=0$)
Bank rotation arc showing $L \cos(\beta)$ and $L \sin(\beta)$ components
(label each component explicitly)

Draw in PowerPoint — use Hicks Fig. 3.x as reference



Gravity in Vehicle-Pointing Coordinates

- ▶ Spherical Earth, inverse-square: $\vec{F}_g = -(\mu m / r^2) \hat{r}$
- ▶ In vehicle-pointing frame:
 - • Radial component: $-\mu m / r^2$
 - • Tangential components: zero (gravity purely radial for spherical Earth)
- ▶ Note: oblateness (J_2) would add a small tangential component — see Appendix
- ▶ Effective gravity in trajectory equations:
 - • $g(r) = \mu / r^2 = g_0 (R_E/r)^2$
 - • At $h=120$ km: $g \approx 9.42 \text{ m/s}^2$ (vs. $g_0 = 9.807 \text{ m/s}^2$ at surface)
 - • Variation over trajectory: $\sim 6\%$ — often approximated as constant g_0
- ▶ The g term in the equations will appear as $-g \sin\gamma$ in $\dot{v}$, $-g \cos\gamma/v$ in $\dot{\gamma}$
- ▶ Centrifugal relief: at circular orbit speed, gravity exactly balances centripetal
 - • Equivalently: when $v^2 / r = g$, the $\dot{\gamma} = 0$ condition is satisfied (orbital case)



Rotating Frame: Coriolis & Centripetal Accelerations

- ▶ When writing EOM in the vehicle-pointing (rotating) frame, e
- ▶ Centripetal acceleration: $-\omega_E^2 \vec{r}_\perp$ (directed away from
- ▶ Coriolis acceleration: $-2 \omega_E \times \vec{v}$
- ▶ In vehicle-pointing components (keeping only first-order in ω
 - Along velocity: $-2 \omega_E [v \cos\gamma \cos\psi \sin\phi - v \sin\gamma \cos\phi \sin\psi]$
 - Vertical plane: $-2 \omega_E v \cos\gamma \cos\psi \cos\phi + \omega_E^2 r \cos\phi \sin\phi$
 - Out-of-plane: $+2 \omega_E v \cos\gamma \sin\psi \cos\phi - 2 \omega_E v \sin\gamma \sin\phi$
- ▶ For many reentry trajectories these terms are small:
 - Coriolis: $\sim 2 \omega_E v \approx 2(7.3e-5)(7000) \approx 1 \text{ m/s}^2$ vs. aero decel > 1
 - Usually kept for accuracy but dropped for analytic solutions
- ▶ Source: Lesson 03-04 — Reentry EOM Development

[ANIMATION / IMAGE]

Rotating Earth view — two ballistic trajectories from same launch point:
Left: non-rotating Earth — lands on target
Right: rotating Earth (Coriolis) — lateral miss by ~350 km at 1000 km range
(use STK / MATLAB or pull from textbook figure)

Generate with MATLAB ode45 or AGI STK screenshot

DISCUSSION | At what downrange distance does Coriolis become operationally significant (>1% effect on landing accuracy)? How would you design a flight test to isolate and measure it?



Deriving $\dot{v}$ — The Speed Equation

- ▶ Project Newton's law onto the velocity direction $\hat{v}$:
 - $m \dot{v} = (\text{along } \hat{v}): -D - mg \sin\gamma - \text{Coriolis}_v$
- ▶ Full form (rotating Earth):

$$\dot{v} = -D/m - g \sin\gamma + \omega_E^2 r \cos\phi (\sin\gamma \cos\phi - \cos\gamma \sin\phi \cos\psi) + 2\omega_E v \cos\gamma \sin\psi \cos\phi$$

- ▶ Physical interpretation.

$$\dot{v} = -(\rho v^2 S C_D)/(2m) - g \sin\gamma \quad [\text{simplified, } \omega_E = 0]$$

- Negative $\gamma \rightarrow -g \sin\gamma > 0 \rightarrow$ gravity aids deceleration for shallow descents
 - Steep descent $|\gamma| \rightarrow 90^\circ$: gravity term $\approx \pm g$
- ▶ Note: v here is relative to atmosphere — must account for ω_E

[DIAGRAM]

Velocity direction projection on vehicle-pointing axes:
Velocity vector, gravity component $-g \sin(\gamma)$ along $\hat{v}$
Drag $-D$ along $\hat{v}$
(label magnitudes, show γ angle)

Draw in PowerPoint — simple free-body diagram



Deriving $\dot{\gamma}$ — The Flight-Path Angle Equation

- ▶ Project Newton's law onto the local-vertical direction (perpendicular to v in vertical plane):
 - $m v \dot{\gamma} = L \cos\beta - mg \cos\gamma + m v^2 \cos\gamma / r + \text{Coriolis}_\gamma$
- ▶ Full form (rotating Earth):

$$\dot{\gamma} = L \cos\beta / (mv) - g \cos\gamma / v + v \cos\gamma / r + 2\omega_E \sin\psi \cos\phi + \omega_E^2 r \cos\phi (\cos\gamma \cos\phi + \sin\gamma \sin\phi \cos\psi) / v$$

- ▶ Physical interpretation.

$$\dot{\gamma} = (\rho v S C_L \cos\beta) / (2m) - (g/v - v/r) \cos\gamma \quad [\text{simplified, } \omega_E = 0]$$

- $v \cos\gamma / r$: centrifugal effect — at orbital speed, balances gravity exactly
 - When $L \cos\beta = 0$ and $v^2 \cos\gamma = g r \cos\gamma$: $\dot{\gamma} = 0 \rightarrow$ circular orbit condition
- ▶ This equation governs whether vehicle skips, glides, or steepens

[DIAGRAM]

Vertical plane free-body:
 $L \cos(\beta)$ upward (blue), gravity $-g \cos(\gamma)/v$ downward (green)
Centrifugal term v^2/r upward (orange)
(balance showing orbital condition when $v^2/r = g$)

Draw in PowerPoint — vertical plane cross-section



Deriving $\dot{\psi}$ — The Heading Angle Equation

- ▶ Project Newton's law onto the out-of-vertical-plane direction:
 - $m v \cos\gamma \dot{\psi} = L \sin\beta + \text{Coriolis}_{\psi} + \text{curvature}_{\psi}$

- ▶ Full form (rotating Earth):

$$\dot{\psi} = L \sin\beta / (m v \cos\gamma) - (v/r) \cos\gamma \tan\phi \sin\psi + 2\omega_E (\cos\phi \tan\gamma \cos\psi - \sin\phi) + \omega_E^2 r \sin\phi \cos\phi \sin\psi / (v \cos\gamma)$$

- ▶ Physical interpretation.

$$\dot{\psi} = (\rho v S C_L \sin\beta) / (2m \cos\gamma) - (v/r) \cos\gamma \tan\phi \sin\psi \quad [\text{simplified, } \omega_E = 0]$$

- This term causes heading drift even without any aerodynamic side force
- ▶ $\beta = 0$ or π : $\dot{\psi} = 0$ in-plane $\rightarrow$ bank modulation controls heading
- ▶ For planar (2-DOF) reduction: set $\dot{\psi} = 0$, $\cos\gamma \tan\phi \sin\psi = 0 \rightarrow$ eliminates this equation



The Complete 3-DOF Equations of Motion ★

- ▶ The full 6-equation system (non-rotating Earth, no thrust baseline):

- $\dot{r} = v \sin\gamma$
- $\dot{\phi} = (v \cos\gamma \cos\psi) / r$
- $\dot{\lambda} = (v \cos\gamma \sin\psi) / (r \cos\phi)$
- $\dot{v} = -(\rho v^2 S C_D)/(2m) - g \sin\gamma$
- $\dot{\gamma} = (\rho v S C_L \cos\beta)/(2m) - (g/v - v/r) \cos\gamma$
- $\dot{\psi} = (\rho v S C_L \sin\beta)/(2m \cos\gamma) - (v/r) \cos\gamma \tan\phi \sin\psi$
-
-
-
-

This is the governing system for ALL analytic solutions and numerical simulations in Part 3

TPS MOMENT | These 6 equations ARE the flight vehicle's mathematical flight manual. Every test limit, every corridor boundary, every guidance law constraint in Part 3 flows from exactly this system. PPT TIP: Animate each equation line to appear on click for maximum impact.